A sharper regular-shell enclosure and correlated initial-data errors

Ricardo Maldonado
Independent Researcher
Henderson, Nevada, United States

23 October 2026
Conditional mathematical working note

Abstract

For one fixed Einstein–scalar model at detuning $\delta = 10^{-3}$, an exact rational residual comparison sharpens the retained regular branch's shell scalar enclosure to radius 9.505×10^{-18}. A polynomial graph and an explicit damping barrier retain the regular-cone uncertainty without numerical shooting. The same shell parameter supplies a separate initial-data calculation using paired Taylor jets and exact difference identities. This finite checkpoint inherits its branch and field-equation premises; it evaluates neither the full second-order response nor an invariant observable.

1 The retained model and branch

Set

$$W(\eta) = \frac{1}{3} + \eta^2 + \frac{\eta^3}{3}, \quad f(\eta) = 1 + c + c\eta, \quad c = \frac{96714321425692}{161839258930731}, \quad \delta = \frac{1}{1000}, \tag{1}$$

$$U(\eta) = \frac{(W')^2}{2} - \frac{2W^2}{3}, \qquad \sigma = 2W + \delta f. \tag{2}$$

The regular interior has $\rho(0) = 0$, $\rho_y(0) = 1$, $p(0) = 0$, with $p = \eta_y$, $a = \rho_y/\rho$ and shell scalar $\beta = \eta(T)$. The retained theorem gives existence and uniqueness in its specified small Dirichlet tube [1], and $|\beta + (9c/64)\delta| \le 3.196\delta^2$. The new necessary enclosure of that same existing branch is

$$\boxed{-0.00008405268308665527 \le \beta \le -0.00008405268308663626}, \quad \mathrm{rad}(\beta) = 9.505 \times 10^{-18}. \tag{3}$$

The endpoints are exact rationals. The old radius is 3.196×10^{-6}. Neither enclosure denotes measured physical precision. This calculation does not independently construct an endpoint solution or enlarge the old hypotheses.

Write $h = \rho^{-2}$, the quantity also denoted H^2, and $z = p/\eta$. Along this negative scalar branch,

$$\eta_y = \eta z, \quad h_y = -2ah, \quad z_y = M(\eta) - 4az - z^2, \quad a^2 = h + \frac{\eta^2 z^2}{12} - \frac{U(\eta)}{6}, \tag{4}$$

where $M = U'/\eta$ extends polynomially to zero. Convexity and the scalar divergence equation give $\eta, p < 0$ and $|\eta(y)| \le |\beta| < \epsilon := 9 \times 10^{-5}$. The cone expansion gives $z \to 0$. Exact polynomial bounds give $0 < M < 4$ and $U(0) \le U < 0$. The constraint preserves the positive branch of a; a first-crossing

test then gives $0 \le z \le 2$. The decreasing h crosses $h_0 = 10^{-2}$ before the shell. On the remaining segment the constraint gives

$$a_- := \frac{1}{9} - 10^{-6} \le a \le a_+ := \frac{3}{20}. \tag{5}$$

No tiny center amplitude or numerical cone-start data need be resolved.

2 A polynomial defect and its damping barrier

The supplement lists exact rational degree-four polynomials $Z(\eta, h)$ and $A(\eta, h)$, beginning with $Z = 14/9 - (42/5)h + (25/23)\eta + \cdots$ and $A = 1/9 + (9/2)h - (7/27)\eta^2 + \cdots$. Define

$$\begin{aligned} B &= 2hZ_h - 4Z, \qquad R = M - Z^2 - \eta Z_\eta Z + AB, \\ D &= h + \eta^2 Z^2/12 - U/6 - A^2. \end{aligned} \tag{6}$$

All coefficients of R, D through total degree four vanish exactly. The remaining coefficients are bounded, not discarded. On $|\eta| \le \epsilon$, $0 \le h \le h_0$, coefficientwise rational estimates prove $0 < Z < 2$, $A > 0$, $B < 0$, and

$$4a_- + \inf Z - \sup|\eta Z_\eta| > 1.88625 \ge \kappa := 37/20. \tag{7}$$

No invariant manifold or convergence of a formal series is assumed. For $e = z - Z(\eta, h)$ along the actual trajectory, the exact identities are

$$\begin{aligned} a - A &= \frac{D + \eta^2(z+Z)e/12}{a + A}, \\ e_y &= R + \frac{BD}{a+A} - \mathcal{K}e, \qquad \mathcal{K} = 4a + z + Z + \eta Z_\eta - \frac{B\eta^2(z+Z)}{12(a+A)} \ge \kappa. \end{aligned} \tag{8}$$

The square-root defect is charged. The final term in $\mathcal{K}$ is nonnegative because $B < 0$ and $z + Z > 0$.

Let $Q = \sup|B|/(a_- + \inf A)$. Absolute coefficients in R and QD, with $|\eta|^i \le \epsilon^i$ and $h^j \le h_0^{j-5} h^5$ for $j > 5$, give nonnegative rational C_j satisfying $|R + BD/(a+A)| \le \sum_{j=0}^{5} C_j h^j$. Use

$$E(h) = 2(h/h_0)^6 + \sum_{j=0}^{5} \frac{C_j h^j}{\kappa - 2ja_+}. \tag{9}$$

The denominators have minimum $\kappa - 10a_+ = 7/20$; the memory term has margin $\kappa - 12a_+ = 1/20$. Since $(h^j)_y + \kappa h^j = (\kappa - 2ja)h^j$, we have $E_y + \kappa E \ge \sum C_j h^j$. At entry $h = h_0$, $|z - Z| \le 2 \le E(h_0)$. Scalar comparison yields $|e| \le E(h)$. The first term retains all cone histories allowed by the entry interval. At the shell, the verified bound is $E(h) < 3.752255 \times 10^{-13}$.

The shell laws retain one common parameter:

$$\begin{aligned} h_s(\beta) &= \delta\left(\frac{Wf}{9} - \frac{W'c}{12}\right) + \delta^2\left(\frac{f^2}{36} - \frac{c^2}{48}\right), \\ p_s &= -W'(\beta) - \delta c/2, \quad a_s = W(\beta)/3 + \delta f(\beta)/6, \quad \rho_s = h_s^{-1/2}. \end{aligned} \tag{10}$$

For $G(\beta) = \beta Z(\beta, h_s(\beta)) + W'(\beta) + \delta c/2$, the shell obeys $|G(\beta)| \le \epsilon E(h_{\text{upper}})$. The full chain derivative $G' = Z + \beta(Z_\eta + Z_h h_s') + W''$ is greater than 3.55469 on the entire old enclosure. Four exact rational Newton proposals select a seed b, supplying no existence assertion. The bound

$$|\beta - b| \le \frac{|G(b)| + \epsilon E(h_{\text{upper}})}{\inf G'}$$

and one outward rational contraction give (3). This is a necessary enclosure using inherited existence, not an independent finite-Dirichlet solve.

3 Paired initial-data differences

The physical initial prescription retains the five means $(K_0, L_0, N_0, J_0, w_0)$ and the lapse, mixed and scalar terms [2]. Their Gram matrix is $\mathrm{diag}(1, 2, 1, 3, 1)$. Put $\ell = 10^{-6}$. Let $\delta_0, \delta_r, \delta_v$ bound errors in the displacement, scaled radial derivative, and ℓ times the ordinary physical-time velocity. Reflected free-wave estimates give

$$e_{\mathrm{free}} = \max\left\{\frac{\delta_0 + (3/2)\delta_v}{3},\ \delta_r + \delta_v\right\} \tag{11}$$

in the norm $\|V\|_X = \max(\sup|V|_H/3, \sup|V_x|_H, \sup|V_r|_H)$ on $D = \{0 \le x \le 3/2,\ \max(0, x - 5/4) \le r \le 7/4 - x\}$. Only L_0 has the Dirichlet row; the other rows use the registered reflection. Compatibility is retained by the exact zero neighborhood of the shell in the initial data. The earlier candidate's certified cap, below 4.561×10^{33}, failed the provisional 10^{28} gate. This was not an actual-error lower bound.

The new evaluator carries actual and nominal jets together with their small differences. For example,

$$AB - \widehat{A}\widehat{B} = (A - \widehat{A})\widehat{B} + A(B - \widehat{B}), \qquad A^{-1} - \widehat{A}^{-1} = -\frac{A - \widehat{A}}{A\widehat{A}}.$$

This avoids charging the full independent variation of both profiles over each cell. All shell inputs derive from the shared β in (10). Their interval projections then feed the jet recurrence; this valid enclosure can lose parameter correlation and is not a full parameter Taylor model.

Degree-four local background polynomials supply jets through the required order. The exact ODE recurrence and a retained whole-footprint enclosure bound normalized fifth derivatives for differentiated Taylor remainders. The registered rational packet, cutoff and exponential approximation have separate direct difference bounds. Paired identities propagate three error channels: shell Taylor-coefficient differences; packet and representation differences; and validated ODE Taylor remainder. Outward arithmetic is included in these channels, with every physical derivative scaling factor retained. Ninety-six closed cells cover $1 \le r \le 7/4$; the extension on $0 \le r \le 1$ is exactly zero.

The registered recentering and its bound. Using the new branch interval with the frozen original candidate gives the still-failing certified cap 1.758×10^{30}. One permitted representation repair replaces only the nominal degree-four background coefficients, chosen from the common certified β midpoint and shell laws with an explicit rational square-root approximation. The packet, cutoff, exponential approximation and physical initial formulas are unchanged. The exact candidate coefficient identity is recorded in the supplement.[1] This is a new explicit candidate; the old one and its failed report remain frozen. For all 96 cells, the three-channel Gram-norm bounds are as follows; every displayed number is rounded strictly upward from the exact rational certificate.

Contribution	δ_0	δ_r	δ_v
Shell coefficients	1.590×10^{19}	1.085×10^{21}	2.424×10^{11}
Packet and representation	1.932×10^{24}	8.519×10^{27}	4.639×10^{16}
ODE Taylor remainder	2.594×10^{14}	8.806×10^{15}	6.228×10^{6}
Total	1.932×10^{24}	8.519×10^{27}	4.639×10^{16}

The exact ledger, before display rounding, proves

$$e_{\mathrm{free}} < 8.518093 \times 10^{27} < 10^{28}. \tag{12}$$

[1] Canonical coefficient JSON SHA256: `739240283c2bee53afa559c4ce71c1e3356cf2d0ad4e4b0f18b9621c077e20d3`.

Thus the full initial/free inequality passes. The earlier proposed design allocations are not all met: the packet radial cap exceeds its 4×10^{27} allocation and the total radial cap exceeds 8×10^{27}. These are reported upper bounds, not lower bounds on error; acceptance uses the actual combined inequality (11). Values, scaled radial derivatives, ordinary-time velocities, flat extension and the Dirichlet row are checked separately.

4 Verification and limits

The branch authenticates 43 retained payloads plus their manifest against independent external pins and all 44 exact public archive members. The initial calculation separately joins its complete upstream closures and the branch certificate. Normal and optimized Python replays and independent exact checks are recorded in the supplement. Branch arithmetic is rational, with integer square-root bounds. Its finite scope was degree four, one comparison domain, four rational seed steps and one validated contraction. The initial scope permits one registered recentering and the same 96 cells.

Residual validation and polynomial representations are established techniques. Recent examples discuss invariance residuals [3] and polynomial dependence with rigorous remainders [4]; the latter cites high-order Taylor-model integration [5]. These references import no theorem about this shell model. The bounded screening supports no exhaustive coverage or priority claim. Reviews are internal AI-assisted checks, not external peer review or proof-assistant formalization.

Initial precision alone supplies no bulk iteration, evaluated total response, invariant volume sign, nonlinear spacetime family, or evidence for a higher-dimensional origin of the Big Bang. The older source, gauge and manuscript artifacts remain frozen. This working note entails no journal submission.

References

[1] R. Maldonado, *Regular-cone Jacobi identities and a stronger uniform branch estimate*, retained checkpoint (11 October 2026), doi:10.5281/zenodo.23294979.

[2] R. Maldonado, *A causal response estimate and an anchored shell-volume prescription*, retained note (23 October 2026), HDBLAST_TOTAL_RESPONSE_PREFLIGHT_20261023.

[3] O. Hénot and A. Takayasu, *Computer-Assisted Proofs in Dynamical Systems: A Case Study of a Heteroclinic Orbit in the Shimizu–Morioka System* (2026), arXiv:2605.07500v1.

[4] S. Iraky and A. Sogokon, *TERA: A Unified Taylor Model Enabled Reachability Analysis Framework* (2026), arXiv:2607.01189v1.

[5] M. Berz and K. Makino, Verified Integration of ODEs and Flows Using Differential Algebraic Methods on High-Order Taylor Models, *Reliable Computing* **4**, 361–369 (1998), doi:10.1023/A:1024467732637.